

A Combo Connectivity Chipset for IoT Applications

This award-winning ultra-low-power product integrates multiple connectivity methods and is designed in India (though the company offering it is in the USA) exclusively for Internet of Things paradigm

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The Internet of Things (IoT) is basically the connection and communication of diverse devices to each other and to the Internet. The back-end of any IoT device consists of elements like microcontrollers, integrated chips and sensors, while the front end is where the communication and connectivity options come, forming the digital side of device.



Image of actual SoC

While designing an IoT device, a designer has different connectivity options to choose from, depending on several factors like the range of communication and power consumption with some examples being Zigbee, Bluetooth 4.0 and Wi-Fi. However, the communication backbone is the IP based network, and there is a need for gateway devices to bridge data across these protocols. The RS9113 M2MCombo chipset provides this functionality of a gateway device in the most integrated manner in the industry today – resulting in lower cost, greater energy efficiency and higher quality of service (QoS) than others.

This product was designed

and developed by a team of over 70 Indian engineers under N. Venkatesh, senior vice president of advanced technologies at Redpine Signals. Based in San Jose California, Redpine Signals is a wireless systems company founded in 2001. They specialise in chipset and system level products for wireless applications.

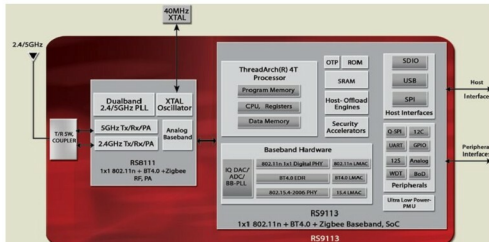
What is it?

The RS9113 is an ultra-low-power, single spatial stream, system on a chip (SoC) that brings together multiple wireless connectivity methods which include 1 Tx-1 Rx dual-band 802.11n Wi-Fi, dual-mode Bluetooth 4.0 and ZigBee (802.15.4-2006) – all these functionalities into a single module. The chipset is a fully self-contained solution, and does not require any part of the protocols it supports to run on a host processor.

This device is basically a complementary metal-oxide-semiconductor (CMOS) chipset that contains a four-threaded processor with on-chip ROM (read-only memory) and RAM (random access memory).

It has integrated baseband digital signal processing, analogue front-end, analogue peripherals, support for digital peripherals, crystal oscillator, dual-band RF transceiver, dual-band high power amplifiers, calibration memory, baluns, diplexers, diversity switch and Quad-SPI Flash.

The processor in the device runs MAC protocol as well as network protocol, enabling the integration of wireless interfaces to embedded systems



Block diagram of SoC